

Project Brief — cikgurohani.com LMS Admin + Parent/Student Portal

1. Project Overview

Project ini ialah sistem LMS ringan untuk cikgurohani.com yang menggabungkan:

1. **Admin Panel** untuk urus pelajar, ibu bapa, pembayaran, kandungan kelas, refund, WABA/WhatsApp, notifikasi, laporan dan tetapan.
2. **Portal Ibu Bapa** untuk semak anak, yuran, transaksi, laporan, bahan kelas, renewal, komen dan notifikasi.
3. **Portal Pelajar** untuk akses bahan kelas, rakaman, Zoom link dan Bank Soalan berdasarkan status langganan.
4. **Payment + Notification Workflow** untuk renewal, pengesahan bayaran, refund, WhatsApp template, email dan portal notification.

Prototype sekarang ialah **frontend-only demo** menggunakan React state dan dummy data. Developer perlu convert logic ini kepada sistem production dengan backend, database, authentication, file storage, real payment gateway, WABA API dan notification service.

2. Main Objective

Membina sistem admin dan portal yang membolehkan cikgurohani:

- Mendaftarkan ibu bapa dan pelajar.
- Mengurus pakej langganan T4, T5 dan T4&5.
- Mengira status langganan secara automatik.
- Mengawal akses portal berdasarkan status bayaran.
- Mengurus kandungan kelas, rakaman, Zoom link, laporan dan Bank Soalan.
- Membolehkan ibu bapa renew langganan melalui manual receipt upload atau online payment.
- Membolehkan admin approve/reject renewal.
- Menghantar notifikasi melalui portal, email dan WhatsApp Business API.
- Mengurus refund dan rekod kewangan.
- Mengawal permission mengikut role admin.

3. User Roles

3.1 Admin Roles

Pemilik

Full access kepada semua module:

- Dashboard
- Pelajar
- Kandungan
- Notifikasi
- WABA
- Pengesahan Bayaran
- Kewangan
- Refund
- Tetapan
- Permission management

Pembantu

Biasanya boleh bantu operasi harian:

- Urus pelajar
- Tambah pelajar
- Kandungan
- Notifikasi
- WABA
- Mohon refund
- Semak profil
- Tidak semestinya boleh proses refund / akses kewangan penuh

Finance

Fokus kepada:

- Kewangan
- Pengesahan bayaran
- Refund
- Laporan kewangan
- CSV/export
- Tidak semestinya boleh edit kandungan atau tetapan

Permission mesti configurable dalam Tetapan.

4. Core Entities / Data Model

Developer perlu sediakan database untuk entity berikut.

4.1 Parent

Fields:

- id, contoh: PAR-00001
- nama
- emel
- telefon
- lokasi
- tarikh_daftar
- duplicate_parent_id, optional
- username_display, optional
- created_at
- updated_at

4.2 Student

Fields:

- id, contoh: STU-00001
- parent_id
- nama
- telefon
- tingkatan: T4, T5, T4&5
- pakej: Bulanan, 3 Bulan, 6 Bulan
- tarikh_mula
- tarikh_tamat
- saluran_bayaran: BCL, BayarCash, Manual, Pindahan Bank
- aktif: boolean
- created_at
- updated_at

4.3 Package

Package logic:

For T4 / T5:

- Bulanan: RM80, 1 bulan
- 3 Bulan: RM230, 3 bulan
- 6 Bulan: RM450, 6 bulan

For T4&5:

- Bulanan: RM160, 1 bulan
- 3 Bulan: RM460, 3 bulan
- 6 Bulan: RM900, 6 bulan

4.4 Payment / Transaction

Fields:

- id / reference_no
- parent_id
- student_id
- tarikh
- pakej
- jumlah
- saluran
- ref
- resit_url
- refunded_by_refund_id, optional
- status: Berjaya, Menunggu, Ditolak, Refunded
- created_at

4.5 Renewal Request

Fields:

- id, contoh: RNW-0007
- parent_id
- student_id
- pakej
- jumlah
- resit_url
- status: Menunggu, Diluluskan, Ditolak
- sumber: Portal, BayarCash, Manual
- sebab_tolak
- lulus_oleh
- lulus_masa
- created_at

4.6 Package Change Request

Fields:

- id, contoh: REQ-0004
- parent_id
- student_id
- dari_pakej

- ke_pakej
- status: Baharu, Selesai, Ditolak
- created_at

4.7 Profile Change Request

Fields:

- id, contoh: PRF-0001
- parent_id
- student_id
- field
- old_value
- new_value
- status: Baharu, Selesai, Ditolak
- requested_by
- reviewed_by
- created_at

4.8 Refund

Fields:

- id, contoh: RFD-0003
- parent_id
- student_id
- payment_ref
- jumlah
- sebab
- kaedah: Akaun bank, QR DuitNow
- akaun / qr_url
- status: Dimohon, Selesai
- requested_by
- processed_by
- resit_refund_url
- notified: boolean
- created_at
- completed_at

4.9 Content

Separate tables or unified content table:

- bahan kelas
- rakaman
- bank soalan
- pengumuman
- laporan pelajar

- Zoom links

4.10 Notification

Fields:

- id
- parent_id
- student_id, optional
- channel: Portal, Email, WhatsApp
- type
- message
- status: Berjaya, Gagal, Menunggu
- baca: boolean, for portal notification
- provider_response
- created_at

4.11 Activity Log

Fields:

- id
- parent_id
- student_id, optional
- type: log_masuk, pakej, notifikasi, edit, refund
- message
- actor
- internal_note
- created_at

5. Subscription Status Logic

Status pelajar dikira berdasarkan aktif dan tarikh_tamat.

Rules:

1. Jika aktif = false
2. Status: Disekat
3. Portal access locked.
4. Parent/student nampak mesej akaun disekat dan CTA hubungi admin.
5. Jika aktif = true dan tarikh_tamat < today

6. Status: **Tamat**

7. Parent masih boleh login.

8. Student content access perlu limited/locked.

9. Renewal popup akan muncul.

10. Zoom link locked.

11. Parent boleh renew.

12. Pelajar tidak boleh renew sendiri, hanya diarahkan minta ibu bapa renew.

13. Jika **aktif = true** dan baki hari **<= 7**

14. Status: **Akan Tamat**

15. Dashboard highlight.

16. Trigger reminder boleh dihantar.

17. Access masih dibenarkan.

18. Jika **aktif = true** dan baki hari **> 7**

19. Status: **Aktif**

20. Full access ikut pakej.

6. Package Access Logic

6.1 Bahan Kelas

Semua pakej boleh akses bahan kelas selagi status bukan **Disekat**. Untuk status **Tamat**, developer perlu decide sama ada:

- View limited + renewal CTA, atau
- Full lock selepas expiry.

Recommended:

- Expired users boleh lihat halaman utama dan yuran.
- Bahan, rakaman, Zoom dan Bank Soalan locked sehingga renewal.

6.2 Rakaman

Akses ikut tingkatan:

- T4 nampak rakaman T4 dan "Kedua-dua".
- T5 nampak rakaman T5 dan "Kedua-dua".

- T4&5 nampak T4, T5 dan “Kedua-dua”.

6.3 Zoom Link

- Aktif / Akan Tamat: boleh klik join Zoom.
- Tamat: locked dengan mesej renew.
- Disekat: overlay blocked.

6.4 Bank Soalan

Access rule:

- Bulanan: tiada akses Bank Soalan.
 - 3 Bulan: ada akses.
 - 6 Bulan: ada akses.
 - T4&5: ikut folder T4 dan T5.
 - Jika parent pada pakej Bulanan nampak pengumuman Bank Soalan, sistem paparkan upsell CTA “Mohon Tukar Pakej”.
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7. Admin Module Requirements

7.1 Dashboard

Dashboard perlu paparkan:

- Total pelajar aktif.
- Pendapatan bulan ini, jika role ada permission kewangan.
- Jika tiada permission kewangan, paparkan notifikasi gagal.
- Bilangan pelajar tamat dalam 7 hari.
- Kadar renewal.
- Bilangan pelajar aktif T4, T5, T4&5.
- Bilangan notifikasi dihantar hari ini.
- Chart agihan pakej.
- Chart revenue / pelajar baru ikut permission.
- Senarai pelajar perlu perhatian: status **Akan Tamat** dan **Tamat**.
- Log notifikasi terkini.
- Alert jika ada renewal / package request / profile request menunggu semakan.

Scenario:

- Jika tiada pelajar hampir tamat, show empty state.
 - Jika ada notifikasi gagal, highlight merah/amber.
 - Jika role tidak boleh sahkan bayaran, jangan tunjuk CTA pengesahan.
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7.2 Student List

Features:

- Search by nama pelajar, ID pelajar, nama parent, email parent.
- Filter by status: Aktif, Akan Tamat, Tamat, Disekat.
- Filter by tingkatan: T4, T5, T4&5.
- Sort by ID, nama, tarikh tamat, hari tinggal.
- Export CSV jika role ada permission.

On row click:

- Open Profil Ibu Bapa.

Scenario:

- Jika no result, show empty state.
 - Jika user tiada permission `pelajar`, page tidak accessible.
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7.3 Parent Profile

Tabs: 1. Maklumat 2. Sejarah Pembayaran 3. Log Aktiviti 4. Notifikasi 5. Nota

Main actions:

- Edit parent info.
- Edit student info / laras pakej.
- Lanjut langganan.
- Sekat / buka sekatan.
- View/download receipt.
- Request refund.
- Resend failed notification.
- Send WABA template.
- Save internal notes.

Scenario:

- Jika parent ada duplicate flag, show warning untuk semak dan gabungkan.
 - Jika edit Parent ID tetapi ada pending refund/renewal/package request, block perubahan ID.
 - Jika tukar Parent ID, semua related records perlu migrate: students, logs, refunds, renewals, package requests, portal notifications, usernames, payments.
 - Jika notification gagal, admin boleh click "Hantar semula" dan status jadi berjaya.
 - Nota hanya internal untuk admin.
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7.4 Add New Student

Form sections: 1. Maklumat ibu bapa

- Nama
- Telefon
- Email
- Lokasi
- Maklumat pelajar
- Nama
- Telefon
- Tingkatan
- Pakej
- Pembayaran
- Tarikh mula
- Kaedah / saluran
- Resit upload
- Nota

System logic:

- Auto-generate Parent ID dan Student ID.
- Tarikh tamat dikira berdasarkan tarikh mula + package duration.
- Harga dikira berdasarkan tingkatan dan pakej.
- On save:
- Create parent.
- Create student.
- Create payment record.
- Send registration email.
- Create portal notification.
- Add activity log.

Scenario:

- Required: parent name, parent phone, parent email, student name.
- Jika parent sedia ada, developer patut support “attach student to existing parent” pada production.
- Jika email invalid, block submission.
- Jika upload receipt missing for manual payment, status boleh jadi **Menunggu** atau admin boleh force approve bergantung policy.

7.5 Laras / Edit Student Package

Admin boleh update:

- Student ID
- Nama
- Telefon
- Tingkatan
- Pakej
- Tarikh tamat
- Active status

Logic:

- Jika Student ID berubah, update all related renewal records.
- Add activity log.
- Jika pakej berubah dari Bulanan kepada 3/6 Bulan, Bank Soalan access terbuka.
- Jika action datang daripada package request, mark request as **Selesai** dan notify parent.

Scenario:

- Jika nombor telefon student berubah melalui admin, terus berkuat kuasa.
- Jika nombor telefon student berubah melalui portal, perlu approval admin dahulu kerana nombor telefon ialah login key.
- Jika package downgrade dari 3/6 bulan ke Bulanan, Bank Soalan access ditarik balik.
- Jika tarikh tamat diletakkan sebelum hari ini, status automatik jadi **Tamat**.

7.6 Manual Extend Subscription

Admin boleh lanjutkan langganan secara manual.

Fields:

- Pakej
- Tarikh mula langganan
- Tarikh tamat auto-calculate

Rules:

- Untuk T4&5, guna harga dan duration T4&5.
- Jika student sedang disekat, admin mesti tick confirmation bahawa tindakan ini akan buka semula akses.
- On confirm:
- Student package updated.
- Expiry updated.

- Active set true.
 - Activity log created.
 - Jika dari renewal approval, renewal status set **Diluluskan**.
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7.7 Block / Unblock Student

Block action:

- Admin mesti isi sebab dalaman.
- Student **aktif = false**.
- Pending renewal untuk student itu auto-reject.
- Parent terima portal notification generic: akses disekat sementara, hubungi admin.
- Sebab dalaman hanya masuk activity log, jangan expose kepada parent.

Unblock action:

- Student **aktif = true**.
- Parent receive notification akses dibuka semula.
- Activity log created.

Scenario:

- Jika student disekat, portal show overlay locked.
 - Jika ada pending renewal semasa block, renewal status jadi **Ditolak**.
 - Parent tidak nampak sebab dalaman.
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8. Content Management Module

Tabs: 1. Bahan Kelas 2. Rakaman 3. Bank Soalan 4. Pengumuman 5. Laporan Pelajar 6. Zoom Link

8.1 Bahan Kelas

Admin boleh:

- Upload PDF.
- Set target: T4, T5, Kedua-dua.
- Edit item.
- Delete item.
- Optional toggle: auto-create announcement after upload.

Scenario:

- Jika toggle announcement ON:

- System create announcement draft.
- Admin boleh edit sebelum publish.
- Optional WABA blast.
- Jika OFF:
- Upload sahaja, toast success.

Production requirement:

- Actual file upload to storage.
- Store file URL, file size, MIME type.
- Virus/file type validation.
- Access control by student package/status.

8.2 Rakaman

Admin boleh:

- Add recording title.
- Set URL, contoh YouTube link.
- Set target T4/T5/Kedua-dua.
- Edit/delete.
- Optional announcement after upload.

Scenario:

- Jika URL invalid, block save.
- Jika recording untuk T5, T4-only student tidak nampak.
- T4&5 student nampak both T4 and T5.

8.3 Bank Soalan

Bank Soalan menggunakan folder-style navigation.

Initial structure:

- Tingkatan 4
- Topikal
- Ujian Percubaan
- Soalan Sebenar SPM
- Tingkatan 5
- Topikal
- Ujian Percubaan
- Soalan Sebenar SPM

Admin boleh:

- Create folder.
- Reorder folder.
- Upload PDF into selected folder.
- Preview PDF.
- Download PDF.
- Delete file.
- Optional announcement after upload.

Access:

- Only package 3 Bulan and 6 Bulan.
- Bulanan sees upsell.
- T4 sees T4 folder.
- T5 sees T5 folder.
- T4&5 sees both.

Scenario:

- Jika admin upload soalan tanpa pilih folder, block upload.
 - Jika folder kosong, show empty state.
 - Jika file deleted, remove from portal immediately.
 - If announcement ON, create draft announcement with CTA to Bank Soalan.
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8.4 Pengumuman

Admin boleh:

- Create announcement.
- Edit announcement.
- Delete announcement.
- Set audience: Semua, T4, T5.
- Add CTA type:
 - zoom
 - bahan
 - rakaman
 - bank

Portal behavior:

- Announcement filtered by student tingkatan.
- If CTA exists, button navigates to correct tab.
- If CTA bank and student package Bulanan, show upsell instead.

Comments:

- Parent/student can comment on announcement.
- Admin can reply.
- Reply appears in portal thread.

Scenario:

- Empty comment cannot be submitted.
 - Admin reply on Enter should publish.
 - Comments should be linked to announcement ID.
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8.5 Laporan Pelajar

Admin boleh publish monthly report by tingkatan.

Fields:

- Tingkatan: T4 / T5
- Bulan
- Ringkasan guru
- Guru
- Tarikh publish

Portal:

- Student T4 sees T4 reports.
 - Student T5 sees T5 reports.
 - Student T4&5 can see both.
 - If no report, show empty message.
-

8.6 Zoom Link

Admin can update:

- T4 Zoom link
- T5 Zoom link

Actions:

- Save link.
- Send announcement about updated Zoom link.

Portal:

- T4 student sees T4 schedule/link.
- T5 student sees T5 schedule/link.
- T4&5 sees both.
- Expired student sees locked message.
- Disekat student sees blocked overlay.

Production option:

- Can be simple static link management.
 - Optional Zoom API integration not required unless auto-create meeting is needed.
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9. Parent / Student Portal

9.1 Login

Two login modes: 1. Portal Ibu Bapa

- Email + password.
- Parent dashboard shows all children.
- Portal Pelajar
- No password.
- Login using student phone number OR parent registered phone number.
- If phone matches one student, open that student.
- If parent phone matches multiple students, show child selection.

Production security note:

- Phone-only login is convenient but weak.
- Recommended: OTP via WhatsApp/SMS, or magic link, especially if portal contains private student/payment info.

Scenario:

- If phone not found, show "Nombor tidak dijumpai".
 - If parent has multiple children, user must select account.
 - If student has phone "—", cannot login by student phone.
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9.2 Language

Portal supports:

- BM
- EN

When language changes:

- Current tab should map to equivalent translated tab.
 - Content remains same.
 - UI labels update.
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9.3 Onboarding

For active non-expired students:

- Show 4-step onboarding:
 - Home
 - Class Materials
 - Zoom Link
 - Student Reports

Scenario:

- Skip marks onboarding done.
 - Finish shows success toast.
 - Expired students should not see onboarding; renewal prompt takes priority.
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9.4 Parent Dashboard

Parent sees:

- Welcome message.
- List of children.
- Each child card shows:
 - Nama
 - Tingkatan
 - Pakej
 - Tarikh tamat
 - Status
 - Days left

Click child:

- Open child portal view.
-

9.5 Expired Subscription Flow

If student status **Tamat** :

- Show modal overlay.
- Message: subscription expired.
- Parent can renew.
- Student sees instruction to ask parent to renew.
- Parent may click “Lihat kandungan dahulu” to dismiss overlay temporarily.
- Expired account can only choose:
 - Bulanan
 - 3 Bulan

Reason:

- Prototype restricts expired renewal to 1 or 3 months only.
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9.6 Renew / Pay Fees Flow

Steps: 1. Choose package. 2. Choose payment method:

- Online payment via BayarCash / FPX / DuitNow.
- Manual bank transfer / QR + receipt upload.
- Submit payment.
- Show thank you / pending review.

Manual flow:

- Parent chooses package.
- Parent scans QR / bank details.
- Parent uploads receipt.
- System creates renewal request status **Menunggu**.
- Admin sees in Pengesahan Bayaran.
- Parent sees “Payment is being reviewed”.

Online flow:

- Parent chooses package.
- Redirect / open BayarCash checkout.
- On successful payment callback:
 - System creates renewal record status **Diluluskan**.

- Updates student package and expiry.
- Creates payment transaction.
- Sends portal notification.
- Sends WhatsApp confirmation.
- Adds activity log.

Expiry calculation:

- If current subscription already expired: new base date = today.
- If still active: new base date = current expiry date.
- New expiry = base date + package duration.

Scenario:

- If receipt upload missing, block submit.
- If pending renewal already exists, hide renew button or show pending message.
- If renewal rejected, parent sees rejection banner and can resubmit.
- If online callback fails, do not extend subscription; show payment failed/pending.

10. Payment Approval Module

Admin page: `Pengesahan Bayaran`.

Sections: 1. Renewal payment requests. 2. Package change requests. 3. Profile change requests.

10.1 Approve Renewal

Admin action:

- View receipt.
- Manually check bank account.
- Click “Sahkan & Lanjutkan”.
- Choose package and start date.
- Confirm.

System:

- Renewal status = `Diluluskan`.
- Student package updated.
- Student expiry updated.
- Student active = true.
- WhatsApp notification logged.
- Portal notification created.
- Activity log created.

Scenario:

- If student is blocked, admin must confirm that approval will reopen access.
 - If admin lacks permission `sahkanBayaran`, hide approve/reject buttons.
 - If renewal already approved, do not allow approve again.
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10.2 Reject Renewal

Admin selects rejection reason:

- Tiada rekod pembayaran dalam akaun bank.
- Jumlah tidak sepadan dengan pakej.
- Resit tidak sah / tidak jelas.
- Resit pernah digunakan sebelum ini.

System:

- Renewal status = `Ditolak`.
- Reason stored.
- Portal notification sent.
- Activity log created.
- Parent can submit new receipt.

Scenario:

- Rejected renewal should not update expiry.
 - If new pending renewal exists after rejection, portal should not continue showing old rejection as active issue.
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10.3 Package Change Request

Parent can request package change from portal profile.

System:

- Create request status `Baharu`.
- Add activity log.
- Admin sees in Pengesahan Bayaran.
- Admin can send WABA template to discuss/payment.
- Admin clicks "Laksanakan Tukar Pakej".
- System opens Laras Student prefilled with requested package.
- After admin saves:
- Request status = `Selesai`.
- Student package updated.

- Parent notified.
- Activity log created.

Scenario:

- Package change request alone does not auto-change package.
 - Payment confirmation may be required manually before admin executes.
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10.4 Profile Change Request

Sensitive changes from portal, especially student phone number, need approval.

Flow:

- Parent/student edits phone number.
- System creates profile request.
- Old phone remains active for login.
- Admin approves/rejects.
- If approved:
 - Student phone updated.
 - Portal notification sent.
 - Activity log created.
- If rejected:
 - No data change.
 - Parent notified.

Scenario:

- Parent profile fields like name/email/parent phone may sync immediately.
 - Student phone should not sync immediately because it is login key.
 - Duplicate pending phone change should be prevented or grouped.
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11. Refund Workflow

Refund can be requested from parent payment history.

11.1 Request Refund

Admin selects payment and fills:

- Refund amount
- Reason
- Refund method:
- Bank account

- QR DuitNow upload

System:

- Create refund status `Dimohon`.
- Add refund to Refund page.
- Add activity log.
- Toast/notification to owner.

Scenario:

- Refund amount cannot exceed original payment amount.
 - Cannot request refund if payment already refunded.
 - If QR method selected, QR upload required.
 - If bank method selected, bank details required.
-

11.2 Process Refund

User with permission `prosesRefund` :

- Opens refund page.
- Uploads refund transfer receipt.
- Marks refund as `Selesai`.

System:

- Refund status = `Selesai`.
- Store receipt.
- Mark original payment as refunded.
- Portal notification to parent.
- Activity log created.

After refund complete:

- Admin can view receipt.
- Admin can download receipt.
- Admin can mark "Maklumkan Admin".
- Admin can send WABA message to parent.

Scenario:

- Cannot mark completed without receipt.
 - Refund completion should not automatically cancel student subscription unless business rule is added.
 - Payment history should display `REFUNDED`.
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12. Notification System

Channels:

- Portal notification
- Email
- WhatsApp Business API / WABA

Notification types:

- Peringatan 7 hari
- Peringatan 3 hari
- 3 hari selepas tamat
- Akses tamat
- Pengesahan daftar
- Kandungan baru
- Renewal
- Refund
- Profile change
- Package change
- Account blocked/unblocked

Admin Log Notifikasi:

- Filter by date from/to.
- Filter by channel.
- Filter by trigger type.
- See status success/fail.
- Retry failed notification.

Scenario:

- If notification failed, show red row and "Hantar semula".
- Retry should create provider call again and update status.
- Store provider response/error for debugging.
- Notification log should be immutable or keep retry history in production.

13. WABA / WhatsApp Module

Tabs: 1. Dashboard 2. Device Manager 3. Inbox 4. Template 5. Blaster 6. Contacts 7. File Manager

13.1 Dashboard

Show:

- Messages sent today.

- Delivery rate.
- Failed messages.
- Active templates.
- 7-day message chart.
- Device status and quota.

13.2 Device Manager

Show connected numbers:

- Number name
- Phone number
- Status: connected/disconnected
- Quality rating
- Daily quota

Actions:

- Add number/device.
- Reconnect disconnected device.
- Disconnect active device.

Production:

- Use official Meta WhatsApp Cloud API.
- Device Manager may be replaced by Meta Business integration status.

13.3 Inbox

Features:

- List parent conversations.
- Unread count.
- Message thread.
- Send message if 24-hour customer service window is active.
- Mark unread as read when opened.

Scenario:

- If 24-hour window expired, free-form message blocked; only approved template allowed.
- Store inbound/outbound webhook events.

13.4 Template

Admin can:

- View templates.

- Create template.
- Select category: Utility / Marketing.
- Use variables like `{{1}}`, `{{2}}`.
- Submit for Meta approval.
- Resubmit rejected template.

Rules:

- Only approved templates can be used in blast.
- Marketing template only to opt-in contacts.
- Rejected template should show rejection note.

13.5 Blaster

Admin selects:

- Approved template.
- Audience:
 - Semua
 - T4 sahaja
 - T5 sahaja
- Akan tamat ≤ 7 hari
- Tamat
- Schedule:
 - Sekarang
 - Malam ini 8 PM
 - Esok 9 AM

Scenario:

- If template not approved, cannot select.
- If marketing template, filter only contacts with marketing opt-in.
- Scheduled blast should use queue/worker.
- Log each send attempt.

13.6 Contacts

Contacts sync from parent records. Fields:

- Name
- Phone
- Labels
- Marketing opt-in
- Chat action

13.7 File Manager

Admin can upload media for templates:

- Image
- PDF/document

Production:

- Store media in object storage.
 - Register/upload media to Meta if required.
 - Validate file type and size.
-

14. Finance Module

Dashboard metrics:

- Revenue by selected period.
- MRR.
- Active subscriptions.
- ARPU.
- Renewal rate.
- Churn rate.
- Average subscription duration.
- LTV.
- Revenue chart.
- Revenue by package.
- Payment channel split.
- Recent transactions.

Periods:

- Harian
- Mingguan
- Bulanan
- Suku Tahun
- YTD

Actions:

- Generate PDF report.
- Export CSV.
- View/download receipt.

Production:

- Revenue should be calculated from actual paid transactions.
 - Refunds should reduce net revenue if accounting requires.
 - Define whether MRR is gross or net.
 - Define churn formula clearly.
-

15. Settings Module

Features:

- Manage admin users.
- Add admin user.
- Change role.
- Remove user.
- Permission matrix by role.
- Reset permissions to default.
- Save WhatsApp community invitation link.
- Set default portal language.

Rules:

- Owner/Pemilik must always retain settings permission.
 - Prevent system from having zero owner.
 - Permission changes apply immediately.
 - In production, permission changes must be logged.
-

16. Required Integrations

16.1 Authentication

Required:

- Admin login.
- Parent login.
- Student login or OTP.
- Password reset for parent.
- Session management.
- Role-based access control.

Recommended stack:

- NextAuth, Supabase Auth, Firebase Auth, Clerk, or custom JWT.

16.2 Database

Required:

- PostgreSQL / MySQL / Supabase / Firebase.
- Need relational links between parent, student, payments, renewals, refunds, logs and notifications.

16.3 File Storage

Required for:

- PDF bahan kelas.
- Bank Soalan PDFs.
- Receipt uploads.
- Refund receipts.
- WABA media.
- QR uploads.

Recommended:

- S3-compatible storage, Supabase Storage, Cloudflare R2, Firebase Storage.

16.4 Payment Gateway

Need connect:

- BayarCash for online FPX/DuitNow.
- Webhook/callback endpoint.
- Verify payment signature.
- Store transaction reference.
- Auto-activate subscription on successful callback.

Manual payment:

- QR / bank transfer receipt upload.
- Admin approval required.

16.5 WhatsApp Business API

Need connect:

- Meta WhatsApp Cloud API or approved BSP.
- Template management.
- Message sending.
- Delivery status webhook.
- Inbound message webhook.
- Media upload.
- Opt-in tracking.

16.6 Email

Need connect:

- SMTP / SendGrid / Mailgun / AWS SES / Resend.
- Registration email.
- Password reset email.
- Renewal confirmation.
- Failed payment notice.
- Content update notice.

16.7 Scheduler / Cron Jobs

Required for:

- Daily reminder 7 days before expiry.
- Reminder 3 days before expiry.
- Access tamat notification on expiry date.
- 3 days after expiry follow-up.
- Scheduled WABA blasts.
- Daily finance summary, optional.

16.8 PDF/CSV Export

Need:

- CSV export for student list and transactions.
- PDF report generator for finance reports.
- Receipt PDF generation.

16.9 Optional Zoom Integration

Not mandatory. Current logic only stores Zoom links manually.

Optional:

- Zoom API to auto-create meetings.
- Per-class recurring meeting link.

17. Critical Scenarios / What Should Happen If...

Scenario	Expected System Behavior
Student subscription expires	Status becomes Tamat ; parent sees renewal prompt; Zoom/ content locked based on access policy.

Scenario	Expected System Behavior
Subscription has 7 days or less	Status becomes Akan Tamat ; dashboard highlights; reminder can be sent.
Student is blocked by admin	Status becomes Disekat ; portal overlay locked; pending renewal rejected; parent notified with generic message.
Admin unblocks student	Student active again; parent receives portal notification.
Parent submits manual renewal receipt	Renewal request created as Menunggu ; admin must verify manually.
Admin approves renewal	Student package/expiry updated; renewal status approved; payment/log/notification created.
Admin rejects renewal	Renewal status rejected; parent sees reason and can resubmit.
Parent pays online successfully	BayarCash callback auto-approves; subscription extends immediately.
Payment callback fails	Do not extend subscription; show failed/pending payment status.
Parent requests package change	Request created; admin must execute manually after confirmation/payment.
Parent changes student phone	Create profile request; old phone remains login key until admin approves.
Admin approves phone change	Student phone updated; parent notified.
Admin rejects phone change	No data change; parent notified.
Admin changes Parent ID	Block if pending refund/renewal/request exists; otherwise migrate all related records.
Admin requests refund	Refund record created as Dimohon ; owner/finance sees pending task.
Refund completed	Refund marked Selesai ; original payment tagged refunded; parent notified.
Notification fails	Log as Gagal ; allow resend; store provider error.
WABA template rejected	Cannot use in blast; allow resubmit after edit/fix.
Marketing blast selected	Send only to opted-in contacts.
Monthly package user opens Bank Soalan	Do not show files; show upsell to upgrade.
Content uploaded with announcement toggle ON	Create editable announcement draft before publishing/blast.
File upload invalid	Reject upload with clear error.

Scenario	Expected System Behavior
User role lacks permission	Hide navigation/action and block API access server-side.

18. API Endpoints Needed

Suggested backend endpoints:

Auth

- POST /auth/admin/login
- POST /auth/parent/login
- POST /auth/student/otp/request
- POST /auth/student/otp/verify
- POST /auth/logout
- POST /auth/password-reset

Parents & Students

- GET /parents
- POST /parents
- GET /parents/:id
- PATCH /parents/:id
- GET /students
- POST /students
- PATCH /students/:id
- POST /students/:id/block
- POST /students/:id/unblock

Payments & Renewals

- GET /payments
- POST /renewals/manual
- POST /renewals/:id/approve
- POST /renewals/:id/reject
- POST /payments/bayarcash/create-checkout
- POST /webhooks/bayarcash

Refunds

- GET /refunds
- POST /refunds
- POST /refunds/:id/complete

Content

- GET /content
- POST /content/materials
- POST /content/recordings
- POST /content/question-bank
- POST /content/announcements
- POST /content/reports
- PATCH /content/:id
- DELETE /content/:id

Notifications

- GET /notifications
- POST /notifications/send
- POST /notifications/:id/retry
- PATCH /portal-notifications/:id/read

WABA

- GET /waba/templates
- POST /waba/templates
- POST /waba/send-template
- POST /waba/blast
- GET /waba/inbox
- POST /waba/inbox/:threadId/reply
- POST /webhooks/waba

Settings

- GET /settings/permissions
- PATCH /settings/permissions
- GET /admin-users
- POST /admin-users
- PATCH /admin-users/:id
- DELETE /admin-users/:id

19. Non-Functional Requirements

Security

- All admin actions must check permission server-side.
- File uploads must validate type and size.
- Payment webhooks must verify signature.
- Do not expose internal block/refund reasons to parent.
- Sensitive logs should be admin-only.

- Use HTTPS only.

Auditability

Every important action must create activity log:

- Add/edit student.
- Change package.
- Approve/reject renewal.
- Block/unblock.
- Refund request/complete.
- Profile change approval.
- Notification retry.
- Permission change.

Performance

- Student list should support pagination and search.
- Notification log should be paginated.
- WABA inbox should load by thread.
- Files should use CDN/object storage.

Reliability

- Use queue for WABA/email sending.
- Store failed notification attempts.
- Retry support.
- Webhook idempotency for payment callbacks.

20. Prototype Limitations To Convert For Production

Current JSX prototype:

- Uses React local state only.
- Uses dummy demo data.
- Demo date fixed to 11 June 2025.
- File uploads only capture file name.
- Receipts are simulated.
- WABA sends are simulated.
- Email sends are simulated.
- Payment gateway is simulated except named BayarCash.
- Role switch is demo dropdown, not real login.
- No backend validation.
- No database persistence.
- No real file storage.
- No real webhook processing.

Production must replace all of these with real services.

21. Acceptance Criteria

System is considered ready when:

1. Admin can login with correct role and only see permitted modules.
2. Admin can create parent/student and record payment.
3. Student status updates automatically based on expiry and active flag.
4. Parent can login and view children.
5. Student can login securely and view allowed content.
6. Expired account shows renewal flow.
7. Manual renewal creates pending request.
8. Admin can approve/reject renewal.
9. Online payment callback activates subscription automatically.
10. Bank Soalan access follows package rule.
11. Admin can upload materials, recordings and question bank files.
12. Parent/student portal shows content filtered by tingkatan and package.
13. Admin can create announcements and reply to comments.
14. Refund workflow works from request to completion.
15. WABA template sending and logs work.
16. Notification failure/retry works.
17. Finance dashboard uses real transaction data.
18. CSV/PDF export works.
19. All important actions create activity logs.
20. Permission changes are enforced on both frontend and backend.